

Rainfall Trend Analysis of Tamilnadu (1901-2017)

“A Data Analytics Project”

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EXECUTIVE SUMMARY

This project presents a statistical analysis of annual rainfall in Tamil Nadu using historical rainfall data from the India Meteorological Department (IMD) covering the period 1901–2017. The primary objective of the study is to examine long-term rainfall characteristics, seasonal variability, and rainfall trends through statistical analysis and data visualization. Python was used as the primary programming language for data processing and analysis. Libraries including Pandas, NumPy, Matplotlib, SciPy, and PyMannKendall were utilized to perform descriptive statistical analysis, seasonal analysis, correlation analysis, linear regression, rainfall anomaly analysis, and non-parametric trend analysis. The study found that Tamil Nadu receives an average annual rainfall of approximately 940 mm, with significant year-to-year variability. The Northeast Monsoon contributes to the largest share of annual rainfall. Linear regression indicated only a slight negative trend, while the Mann–Kendall trend test confirmed that there is no statistically significant long-term trend in annual rainfall during the study period. This project demonstrates the practical application of Python programming and statistical methods for analyzing real-world climate data and generating meaningful insights through reproducible data analysis.

1. PROJECT OVERVIEW

Rainfall plays a vital role in agriculture, water resource management, hydroelectric power generation, and environmental sustainability. Understanding long-term rainfall patterns helps governments, researchers, and planners make informed decisions regarding irrigation, drought preparedness, flood management, and climate adaptation. Tamil Nadu has a unique rainfall pattern compared to many other Indian states because it receives a significant portion of its annual rainfall from the Northeast Monsoon. This distinctive climatic characteristic makes the analysis of rainfall trends particularly important for the state. This project aims to analyze the historical rainfall data of Tamil Nadu from 1901 to 2017 using statistical techniques and Python-based data analysis tools. The study examines annual, monthly, and seasonal rainfall characteristics and investigates long-term rainfall trends through descriptive statistics, correlation analysis, linear regression, the Mann–Kendall trend test, Sen's slope estimation, and rainfall anomaly analysis. The project demonstrates how statistical methods and programming can be combined to analyze large climatic datasets efficiently, generate meaningful visualizations, and derive evidence-based conclusions from historical rainfall records.

2. OBJECTIVES

The primary objectives of this project are:

- To analyze the historical annual rainfall data of Tamil Nadu for the period 1901–2017.
- To compute descriptive statistical measures such as mean, median, standard deviation, variance, and coefficient of variation.
- To examine monthly and seasonal rainfall patterns and identify periods of high and low rainfall.
- To investigate the relationship between monthly rainfall and annual rainfall using correlation analysis.
- To assess long-term rainfall trends using linear regression, the Mann–Kendall trend test, and Sen's slope estimation.
- To visualize rainfall characteristics through graphs and rainfall anomaly plots using Python.
- To demonstrate the application of Python and statistical techniques in analyzing real-world climate data.

3. DATASET DESCRIPTION

3.1 Data Source

The rainfall dataset used in this project was obtained from the India Meteorological Department (IMD), the official national agency responsible for meteorological observations and weather-related data in India. The dataset contains historical rainfall records for the state of Tamil Nadu.

3.2 Study Period

The analysis covers a period of 117 years, from 1901 to 2017. This long-term dataset provides sufficient observations to examine rainfall variability, seasonal characteristics, and long-term rainfall trends.

3.3 Dataset Structure

The dataset consists of 117 observations (years) and 19 variables. These variables include: SUBDIVISION – Name of the meteorological subdivision (Tamil Nadu) YEAR – Observation year JAN to DEC – Monthly rainfall (in millimeters) ANNUAL – Total annual rainfall (in millimeters) JF – January–February seasonal rainfall MAM – March–May seasonal rainfall JJAS – June–September seasonal rainfall OND – October–December seasonal rainfall The dataset contained no missing values, ensuring that all statistical analyses were performed on a complete dataset without the need for data imputation or preprocessing.

3.4 Data Preparation

The dataset was imported into Python using the Pandas library. Data quality was verified by checking the dataset dimensions, column names, data types, and missing values before performing statistical analysis. This ensured the reliability and accuracy of the subsequent analyses.

4. TOOLS AND TECHNOLOGIES

The project was carried out using Python and several open-source libraries for statistical analysis, data processing, and visualization. The tools and technologies used are listed below.

Tool / Technology	Purpose
Python 3.9	Primary programming language used for data analysis and statistical computations.
Pandas	Used for data import, cleaning, manipulation, and analysis.
NumPy	Used for numerical computations and array operations.
Matplotlib	Used to create graphs and visualizations, including annual rainfall trends and rainfall anomaly plots.
SciPy	Used for statistical analysis, including linear regression.
PyMannKendall	Used to perform the Mann–Kendall trend test and estimate Sen's slope.
Microsoft Excel	Used for initial data organization, verification, and exporting analysis results.
Visual Studio Code (VS Code)	Integrated Development Environment (IDE) used for writing, executing, and debugging Python programs.

The combination of these tools enabled efficient processing of historical rainfall data, automated statistical analysis, and the generation of high-quality tables and visualizations suitable for reporting and interpretation.

5. METHODOLOGY

The methodology adopted in this project consists of a systematic process of data collection, preparation, statistical analysis, trend analysis, and visualization. The historical rainfall dataset was first imported into Python, where it was examined for completeness and consistency. Since the

dataset contained no missing values, it was directly used for statistical analysis. A combination of descriptive and inferential statistical techniques was employed to understand rainfall characteristics and evaluate long-term rainfall trends. Graphical visualizations were also generated to present the findings in an easily interpretable manner. The major statistical techniques used in this study are described below.

5.1 Descriptive Statistics

Descriptive statistics were computed to summarize the overall characteristics of annual rainfall. The following measures were calculated: Mean Median Minimum Maximum Range Standard Deviation Variance Coefficient of Variation (CV) These statistics provide information about the central tendency and variability of annual rainfall over the study period.

5.2 Monthly and Seasonal Analysis

Monthly rainfall statistics were calculated to examine the distribution of rainfall throughout the year. For each month, the mean rainfall, standard deviation, and coefficient of variation were computed. Similarly, seasonal rainfall totals (JF, MAM, JJAS, and OND) were analyzed to understand the contribution and variability of different climatic seasons.

5.3 Correlation Analysis

Pearson's correlation coefficient was calculated between monthly rainfall and annual rainfall to identify the strength and direction of the relationship. The analysis helps determine which months contribute most strongly to the annual rainfall total.

5.4 Linear Regression Analysis

Simple linear regression was applied to examine the long-term trend in annual rainfall over time. The regression model estimates the relationship between year (independent variable) and annual rainfall (dependent variable). The analysis produced the following measures: Regression slope Intercept Correlation coefficient (R) Coefficient of determination (R^2) p-value These statistics were used to assess the direction and statistical significance of the rainfall trend.

5.5 Mann–Kendall Trend Test

The Mann–Kendall trend test is a non-parametric statistical test used to detect the presence of a monotonic trend in time-series data. Unlike linear regression, it does not require the data to follow a normal distribution and is therefore widely used in hydrological and climatological studies. The test provides the following outputs: Trend direction Z-statistic p-value Kendall's Tau

5.6 Sen's Slope Estimation

Sen's slope estimator was used to quantify the magnitude of the long-term rainfall trend. It estimates the median rate of change in annual rainfall over time and is considered more robust than ordinary least squares regression because it is less affected by extreme values.

5.7 Rainfall Anomaly Analysis

Rainfall anomalies were calculated by subtracting the long-term mean annual rainfall from the annual rainfall of each year. A positive anomaly indicates that the rainfall in a particular year was above the long-term average, whereas a negative anomaly indicates below-average rainfall. Rainfall anomaly plots were generated to visualize wet and dry years throughout the study period.

6. RESULTS AND DISCUSSION

This chapter presents the results obtained from the statistical analysis of the annual rainfall data of Tamil Nadu for the period **1901–2017**. The findings are presented using statistical tables, graphs, and appropriate interpretations. Each analysis is discussed to understand rainfall characteristics, seasonal variations, and long-term rainfall trends.

6.1 DESCRIPTIVE STATISTICS OF ANNUAL RAINFALL

Descriptive statistical measures were computed to summarize the characteristics of annual rainfall over the study period. The results are presented in **Table 6.1**.

Table 6.1 Descriptive Statistics of Annual Rainfall

Statistic	Value
Number of Observations	117
Mean	940.47 mm
Median	944.50 mm
Minimum	318.00 mm
Maximum	1365.30 mm
Range	1047.30 mm
Standard Deviation	168.89 mm
Variance	28523.39 mm ²
Coefficient of Variation	17.96%

Interpretation

The average annual rainfall of Tamil Nadu during **1901–2017** was **940.47 mm**, indicating a moderate level of annual precipitation. The median rainfall (**944.50 mm**) is very close to the mean, suggesting that the annual rainfall distribution is approximately symmetric.

The highest annual rainfall recorded during the study period was **1365.30 mm**, while the lowest was **318.00 mm**, indicating considerable year-to-

year variability. The standard deviation of **168.89 mm** further confirms this variability.

The coefficient of variation (**17.96%**) indicates moderate variability in annual rainfall, suggesting that although annual rainfall fluctuates from year to year, the long-term rainfall pattern remains relatively stable.

6.2 MONTHLY RAINFALL ANALYSIS

Monthly rainfall statistics were computed to examine the distribution and variability of rainfall throughout the year. The mean rainfall, standard deviation, and coefficient of variation (CV) were calculated for each month. The results are presented in **Table 6.2**.

Table 6.2 Monthly Rainfall Statistics

Month	Mean (mm)	Standard Deviation (mm)	Coefficient of Variation (%)
January	23.75	32.06	134.98
February	13.21	19.40	146.87

March	19.47	22.30	114.51
April	44.43	28.02	63.08
May	70.23	31.65	45.06
June	52.11	19.00	36.46
July	71.20	30.94	43.46
August	96.08	37.35	38.87
September	111.52	39.31	35.25
October	181.96	67.49	37.09
November	175.38	90.66	51.69
December	81.15	65.45	80.66

Interpretation

The monthly rainfall analysis reveals a distinct seasonal rainfall pattern in Tamil Nadu. The highest average monthly rainfall was recorded during **October (181.96 mm)**, followed by **November (175.38 mm)**. These two months contribute significantly to the annual rainfall because they coincide with the **Northeast Monsoon**, which is the principal rainy season in Tamil Nadu.

The lowest average rainfall occurred during **February (13.21 mm)**, indicating very dry conditions during the winter months.

The coefficient of variation was highest in **February (146.87%)** and **January (134.98%)**, indicating highly irregular rainfall during these months. In contrast, the monsoon months from **June to October** exhibited comparatively lower coefficients of variation, suggesting more consistent rainfall patterns.

These findings demonstrate that rainfall in Tamil Nadu is highly seasonal, with the majority of precipitation concentrated during the Northeast Monsoon period.

6.3 SEASONAL RAINFALL ANALYSIS

Seasonal rainfall statistics were computed to understand the contribution and variability of rainfall across the four major climatic seasons in Tamil Nadu. The average rainfall, standard deviation, and coefficient of variation (CV) were calculated for each season. The results are presented in **Table 6.3**.

Table 6.3 Seasonal Rainfall Statistics

Season	Mean (mm)	Standard Deviation (mm)	Coefficient of Variation (%)
JF (January–February)	36.96	37.01	100.13
MAM (March–May)	134.12	45.60	34.00

JJAS (June–September)	330.91	70.05	21.17
OND (October–December)	438.48	131.89	30.08

Interpretation

The seasonal rainfall analysis indicates that the **October–December (OND)** season receives the highest average rainfall (**438.48 mm**), making it the most important rainfall season in Tamil Nadu. This is primarily due to the influence of the **Northeast Monsoon**, which contributes a major share of the state's annual rainfall.

The **June–September (JJAS)** season recorded the second-highest average rainfall (**330.91 mm**), representing the contribution of the Southwest Monsoon.

The **January–February (JF)** season received the lowest average rainfall (**36.96 mm**) and exhibited the highest coefficient of variation (**100.13%**), indicating highly inconsistent rainfall during this period.

Among all seasons, **JJAS** had the lowest coefficient of variation (**21.17%**), suggesting that rainfall during the Southwest Monsoon is relatively more consistent compared to the other seasons.

Overall, the seasonal analysis confirms that Tamil Nadu is predominantly dependent on the **Northeast Monsoon** for its annual rainfall.

6.4 WETTEST AND DRIEST YEARS

The years with the highest and lowest annual rainfall were identified to understand the extremes in rainfall during the study period. The results are presented in **Table 6.4**.

Table 6.4 Wettest and Driest Years

Category	Year	Annual Rainfall (mm)
Wettest Year	2005	1365.30
Driest Year	2002	318.00

Interpretation

The analysis shows that **2005** was the wettest year during the study period, with an annual rainfall of **1365.30 mm**, which is substantially higher than the long-term average of **940.47 mm**.

In contrast, **2002** was identified as the driest year, recording only **318.00 mm** of rainfall. This value is considerably lower than the average annual rainfall and reflects a year of severe rainfall deficiency.

The large difference between the wettest and driest years highlights the significant inter-annual variability in rainfall experienced in Tamil Nadu. Such variability can have important implications for agriculture, water resource management, and drought and flood preparedness.

6.5 Correlation Analysis

Pearson's correlation analysis was performed to examine the relationship between monthly rainfall and total annual rainfall. The correlation coefficient (r) was calculated for each month with respect to annual rainfall. The results are presented in **Table 6.5**.

Table 6.5 Correlation between Monthly Rainfall and Annual Rainfall

Month	Correlation Coefficient (r)
January	0.159
February	0.063
March	0.196
April	0.197
May	0.299
June	0.268
July	0.111
August	0.290
September	0.372
October	0.444
November	0.585
December	0.401

Interpretation

The correlation analysis indicates that rainfall during the **Northeast Monsoon months** has the strongest relationship with annual rainfall in Tamil Nadu.

Among all months, **November** exhibited the highest positive correlation ($r = 0.585$), followed by **October** ($r = 0.444$) and **December** ($r = 0.401$). These findings indicate that rainfall received during these months has the greatest influence on the annual rainfall total.

In contrast, **February ($r = 0.063$)** and **July ($r = 0.111$)** showed relatively weak correlations with annual rainfall, suggesting that rainfall during these months contributes less to the overall annual variability.

Overall, the results confirm the dominant role of the **Northeast Monsoon** in determining the annual rainfall pattern of Tamil Nadu.

6.6 Linear Regression Analysis

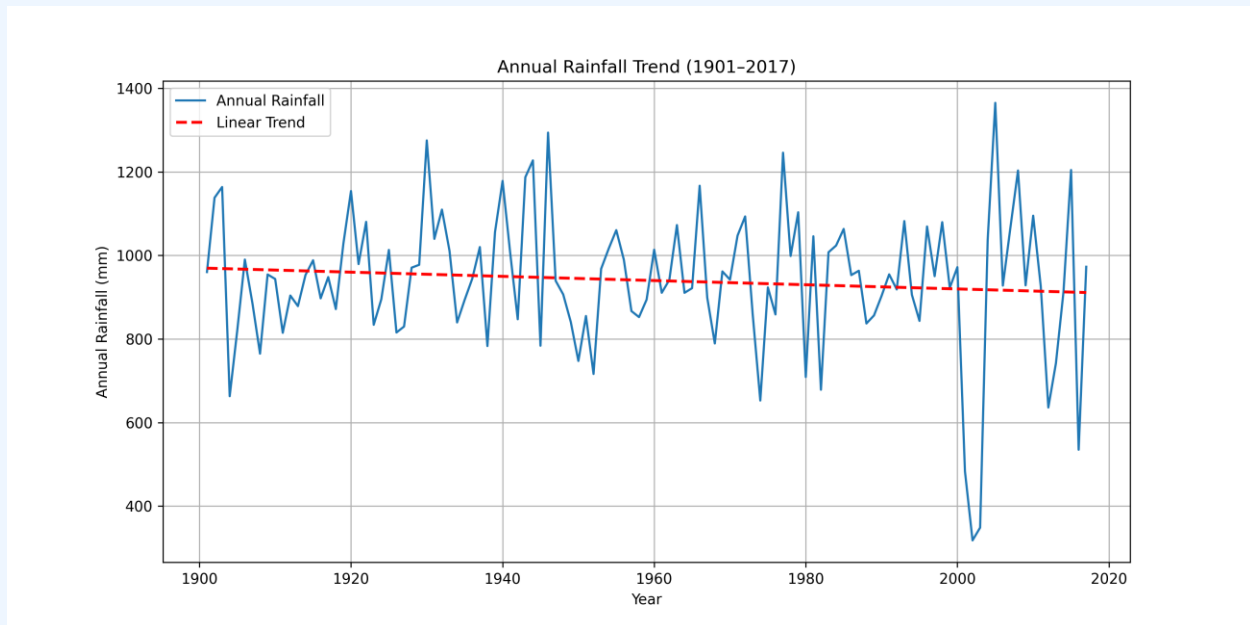
Linear regression analysis was performed to examine the long-term trend in annual rainfall over the study period (1901–2017). The analysis estimated the relationship between year (independent variable) and annual rainfall (dependent variable). The regression results are presented in **Table 6.6**, and the fitted regression line is shown in **Figure 6.2**.

Table 6.6 Linear Regression Results

Statistic	Value
Slope	-0.5021 mm/year
Intercept	1924.16
Correlation Coefficient (R)	-0.1008

Coefficient of Determination (R^2)	0.0102
p-value	0.2793

Figure 6.2 Linear Regression of Annual Rainfall



Interpretation

The regression analysis indicates a **slight negative trend** in annual rainfall over the study period, with a slope of **-0.5021 mm per year**. This suggests that the annual rainfall decreased by approximately **0.5 mm per year** on average.

However, the coefficient of determination ($R^2 = 0.0102$) indicates that only about **1.02%** of the variation in annual rainfall is explained by time. This demonstrates that the relationship between year and annual rainfall is extremely weak.

Furthermore, the **p-value (0.2793)** is greater than the commonly used significance level of **0.05**, indicating that the observed trend is **not statistically significant**.

Therefore, based on linear regression analysis, there is **no strong evidence of a significant long-term increasing or decreasing trend** in the annual rainfall of Tamil Nadu during the period 1901–2017.

6.7 MANN–KENDALL TREND TEST

The Mann–Kendall trend test was performed to determine whether a statistically significant monotonic trend exists in the annual rainfall of Tamil Nadu during the period **1901–2017**. Unlike linear regression, the Mann–Kendall test is a non-parametric method and does not require the data to follow a normal distribution.

The test results are presented in **Table 6.7**.

Table 6.7 Mann–Kendall Trend Test Results

Statistic	Value
Trend	No trend
Hypothesis	Fail to Reject H_0
p-value	0.9343

Z-statistic	0.0824
Kendall's Tau	0.0053
Sen's Slope	0.0450 mm/year
Intercept	941.89

Interpretation

The Mann–Kendall trend test indicates **no statistically significant trend** in the annual rainfall of Tamil Nadu during the study period.

The **p-value (0.9343)** is much greater than the significance level of **0.05**, indicating that the null hypothesis of **no monotonic trend** cannot be rejected. The **Z-statistic (0.0824)** and **Kendall's Tau (0.0053)** are both close to zero, further confirming the absence of a significant increasing or decreasing trend.

The estimated **Sen's slope of 0.0450 mm/year** suggests a very small positive rate of change in annual rainfall. However, this change is negligible and is not statistically significant.

Overall, the Mann–Kendall test supports the findings obtained from the linear regression analysis, indicating that **Tamil Nadu has not**

experienced any significant long-term trend in annual rainfall between 1901 and 2017.

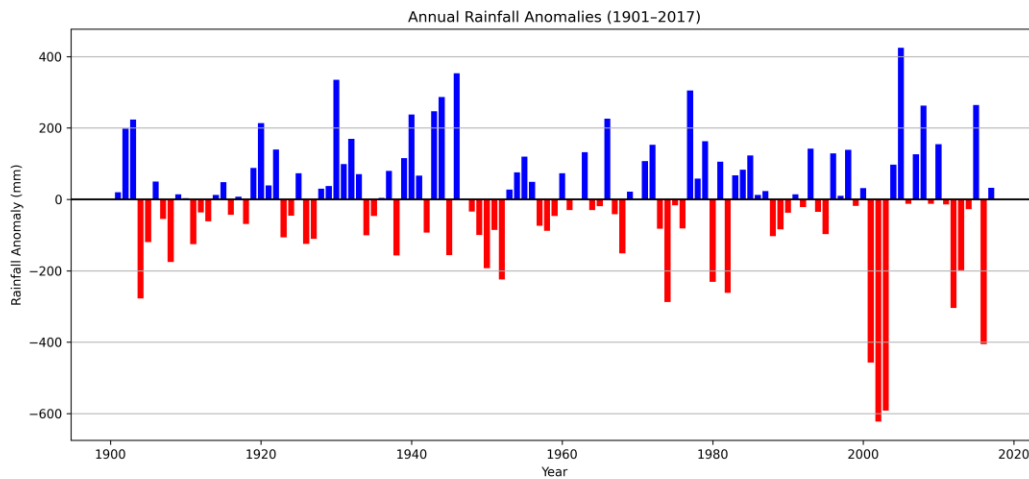
6.8 RAINFALL ANOMALY ANALYSIS

Rainfall anomaly analysis was carried out to examine the variation of annual rainfall from the long-term average rainfall during the study period. A rainfall anomaly represents the difference between the annual rainfall of a particular year and the overall mean annual rainfall.

Positive anomaly values indicate years that received above-average rainfall, while negative anomaly values represent years with below-average rainfall.

The rainfall anomaly plot is presented in **Figure 6.3**.

Figure 6.3 Rainfall Anomaly of Tamil Nadu (1901–2017)



Interpretation

The rainfall anomaly plot illustrates considerable year-to-year fluctuations in annual rainfall throughout the study period. Several years recorded positive rainfall anomalies, indicating above-average rainfall, whereas other years experienced negative anomalies, reflecting below-average rainfall.

The anomalies do not exhibit any sustained increasing or decreasing pattern over time. Instead, the rainfall oscillates around the long-term average, indicating natural climatic variability.

The occurrence of both positive and negative anomalies throughout the study period supports the findings of the linear regression and Mann–Kendall trend analyses, which indicated that there is **no statistically significant long-term trend** in the annual rainfall of Tamil Nadu.

Overall, the rainfall anomaly analysis provides a clear visual representation of inter-annual rainfall variability and demonstrates that extreme wet and dry years occur intermittently rather than following a systematic long-term pattern.

7. CONCLUSION

This project analyzed the historical rainfall patterns of Tamil Nadu using India Meteorological Department (IMD) rainfall data covering the period **1901–2017**. Python-based statistical analysis and visualization techniques were successfully applied to examine annual, monthly, and seasonal rainfall characteristics.

The descriptive statistical analysis showed that Tamil Nadu receives an average annual rainfall of **940.47 mm**, with moderate year-to-year variability. Monthly and seasonal analyses revealed that the **Northeast Monsoon (October–December)** contributes the largest share of the state's annual rainfall, while the winter months receive comparatively little rainfall.

Correlation analysis demonstrated that **November ($r = 0.585$)** exhibited the strongest relationship with annual rainfall, followed by **October ($r = 0.444$)** and **December ($r = 0.401$)**. These findings further confirm the dominant influence of the Northeast Monsoon on Tamil Nadu's annual rainfall.

Linear regression analysis indicated a slight negative slope; however, the trend was not statistically significant. Similarly, the Mann–Kendall trend test detected **no significant long-term monotonic trend** in annual

rainfall during the study period. Rainfall anomaly analysis also showed that fluctuations occurred around the long-term average without any persistent increasing or decreasing pattern.

Overall, the study concludes that although Tamil Nadu experiences considerable year-to-year rainfall variability, there is **no statistically significant long-term trend** in annual rainfall between **1901 and 2017**. The project also demonstrates the effectiveness of Python as a tool for statistical analysis, visualization, and interpretation of large climatic datasets, providing valuable insights for climate studies and water resource planning.

8. FUTURE SCOPE

Although this project provides a comprehensive statistical analysis of historical rainfall in Tamil Nadu, several opportunities exist for further improvement and expansion.

Future work may include:

- Extending the analysis using more recent IMD rainfall data to examine current rainfall trends.
- Applying advanced time-series forecasting techniques such as ARIMA, SARIMA, or machine learning models to predict future rainfall.
- Incorporating additional climatic variables such as temperature, humidity, and atmospheric pressure to study their influence on rainfall patterns.
- Performing district-wise rainfall analysis to identify regional variations within Tamil Nadu.
- Developing an interactive dashboard using tools such as Power BI, Tableau, or Streamlit for real-time visualization and analysis.

- Comparing rainfall trends across different Indian states to understand regional climatic variations.

These extensions would provide deeper insights into rainfall behavior and support better decision-making in agriculture, water resource management, and climate adaptation planning.

9. REFERENCES

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APPENDIX A

Python Source Code

The following Python scripts were developed during this project:

- `read_data.py`
- `descriptive_statistics.py`
- `monthly_statistics.py`
- `seasonal_statistics.py`
- `wet_dry_years.py`
- `correlation_analysis.py`
- `annual_rainfall_plot.py`
- `linear_regression.py`
- `mann_kendall.py`
- `rainfall_anomaly.py`

The complete source code is available in the accompanying GitHub repository.

APPENDIX B

Project Outputs

Excel Files

- Descriptive_Statistics.xlsx
- Monthly_Statistics.xlsx
- Seasonal_Statistics.xlsx
- Correlation_Analysis.xlsx

Figures

- Figure 6.1 Annual Rainfall Plot
- Figure 6.2 Linear Regression Plot
- Figure 6.3 Rainfall Anomaly Plot